Identifying Popular Products at An Early Stage of Sales Season for Apparel Industry

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Introduction

- Traditional rule-based method is both time-consuming and error-prone.
- Research on classifying the popularity of apparel products is limited.

Sell-through Rate	Days on Market					
	<15 days	>= 15 days	>= 30days	>= 45 days	>= 60 days	>= 90 days
< 10%	——	Slow	Slow	Slow	Slow	Slow
>= 10%	——	Slow	Slow	Slow	Slow	Slow
>= 15%	——	Slow	Slow	Slow	Slow	Slow
>= 20%	Average	Slow	Slow	Slow	Slow	Slow
>= 25%	Average	Average	Slow	Slow	Slow	Slow
>= 35%	Fast	Average	Average	Slow	Slow	Slow
>= 45%	Fast	Fast	Average	Average	Slow	Slow
>= 55%	Fast	Fast	Fast	Average	Slow	Slow
>= 65%	Fast	Fast	Fast	Fast	Average	Slow
>= 75%	Fast	Fast	Fast	Fast	Average	Average
>= 80%	Fast	Fast	Fast	Fast	Fast	Fast

Figure: Rule-based Method

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Framework

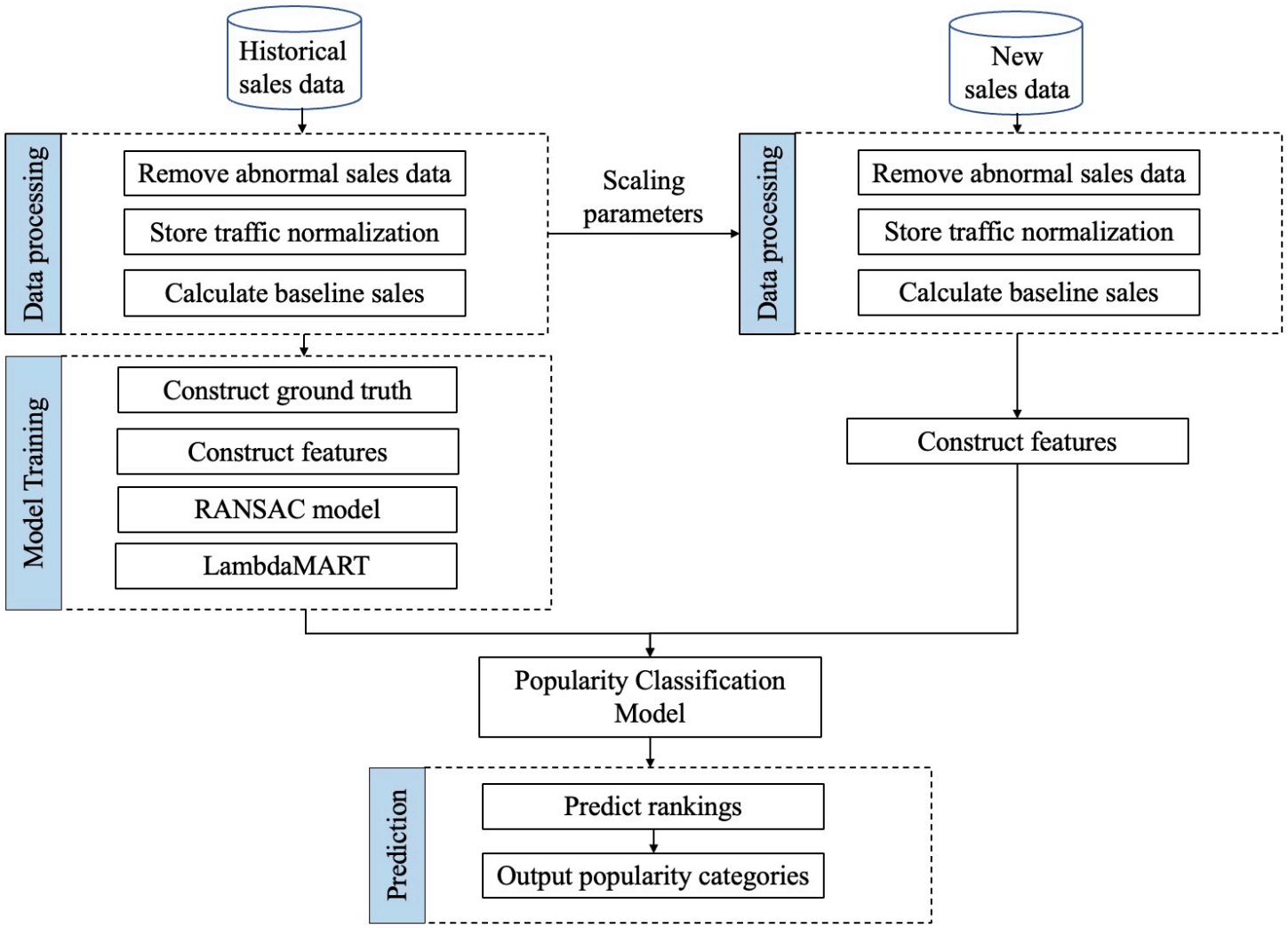


Figure: Framework of the Popularity Classification Model

Data Processing

Data cleaning

Remove sales that are not from individual buyers, including group sales data and returned sales data.

Normalize store traffic

Adjust sales from different stores by normalizing the store traffic in each region to a standardized level.

Baseline sales

Calculate the baseline sales volume using linear regression, which estimates the sales volume that would have occurred if no discounts were applied to the products.

Dependent Variable: AW Sales

Defition of AW Sales

AW Sales is calculated as the store's average weekly sales during its main sales period within a given region, using adjusted sales data.

Main sales period identification

- Calculate the percentage of sales for each week.
- Sort the weekly sales percentages in descending order.
- Compute the cumulative percentage of weekly sales in descending order.
- Identify the weeks in which the cumulative percentage exceeds the threshold as the main selling period.
- If the number of weeks in the main selling period is less than the preset minimum number of weeks, additional weeks are included until the minimum is reached.

Independence from Factors

Differences in store traffic, Number of stores with initial stock, Discounts, Length of time the product has been launched.

Independent Variable: Feature Description.

Table 5 Mathematical Notations

Notation	Definition
k	The index of product.
K	Number of products to be ranked.
N_k	Number of stores with an initial stock of product k.
M_k	Number of weeks since the launch of product k.
s_{mnk}	The adjusted sales volume of product k in store n during week m.
x_{mnk}	A binary variable: 1 if store n sells any quantity of product k in week m, and 0 otherwise.
$\mathcal{D}_{nk}$	$\mathcal{D}_{nk}$ is a sequence of days that represent the ordering times of orders of product k in store n. As an illustration, for product k in store n, suppose that we record sales on day 1 (1 order), day 5 (3 orders), and day 7 (2 orders) after its initial launching. The vector $\mathcal{D}_{nk}$ would be denoted as (1, 5, 5, 5, 7, 7). Therefore, $\mathcal{D}_{nk}[0]$ represents the number of days before the first sale of product k in store n since its launching. Subsequently, the difference between $\mathcal{D}_{nk}[d+1]$ and $\mathcal{D}_{nk}[d]$ represents the number of days that elapsed between two consecutive orders of product k in store n.
d_{nk}	Length of the vector $\mathcal{D}_{nk}$.
$\mathcal{L}_k$	$\mathcal{L}_k$ refers to the longest increasing subsequence derived from the sequence of "weekly adjusted sales volume" of product k across all stores within a typical region. To illustrate, consider product k with a weekly adjusted sales volume sequence of (50, 162, 364, 285, 398, 488) in a region. In this case, $\mathcal{L}_k$ would be (50, 162, 364, 398, 488).
l_k	Length of $\mathcal{L}_k$.

Note. Here, $\mathcal{D}_{nk}[d]$ and $\mathcal{L}_k[d]$ denote the $(d+1)$th elements of $\mathcal{D}_{nk}$ and $\mathcal{L}_k$, respectively.

Figure: Mathematical Notations

Table 6 Definitions of Variables Used in the Popularity Classification Model

Variable	Definition
Mean sales	$\sum_{n=1}^{N_k} \frac{\sum_{m=1}^{M_k} s_{mnk}}{M_k} \cdot \frac{1}{N_k}$
Mean maximal sales	$\sum_{n=1}^{N_k} \max_m(s_{mnk}) \cdot \frac{1}{N_k}$
Active store ratio	$\sum_{n=1}^{N_k} \frac{\sum_{m=1}^{M_k} x_{mnk}}{M_k} \cdot \frac{1}{N_k}$
First sale waiting days	$\sum_{n=1}^{N_k} \mathcal{D}_{nk}[0] \cdot \frac{1}{N_k}$
Mean waiting days	$\sum_{n=1}^{N_k} \sum_{d=0}^{d_{nk}-1} \frac{\mathcal{D}_{nk}[d+1] - \mathcal{D}_{nk}[d]}{d_{nk}-1} \cdot \frac{1}{N_k}$
Increasing subsequence length (Length)	l_k
Increasing subsequence range (Range)	$\mathcal{L}_k[l_k - 1] - \mathcal{L}_k[0]$
Increasing subsequence growth rate (Growth Rate)	$\frac{\mathcal{L}_k[l_k - 1] - \mathcal{L}_k[0]}{l_k}$

Figure: Definitions of Variables

Prediction Model: Ranking-based Algorithm.

Evaluation metric

$$DCG = \sum_{i=1}^{K} \frac{2^{aws_i} - 1}{log(1+i)} \tag{1}$$

$$NDCG = \frac{DCG}{IDCG} \tag{2}$$

In our context, K represents the number of products to be ranked, while aws_i denotes the value of *AW Sales* of the product at position i.
The term *IDCG* refers to the Ideal Discounted Cumulative Gain, which represents the DCG value of the ideal order. The ideal order can be obtained from the *AW Sales* calculated based on the entire sales season.

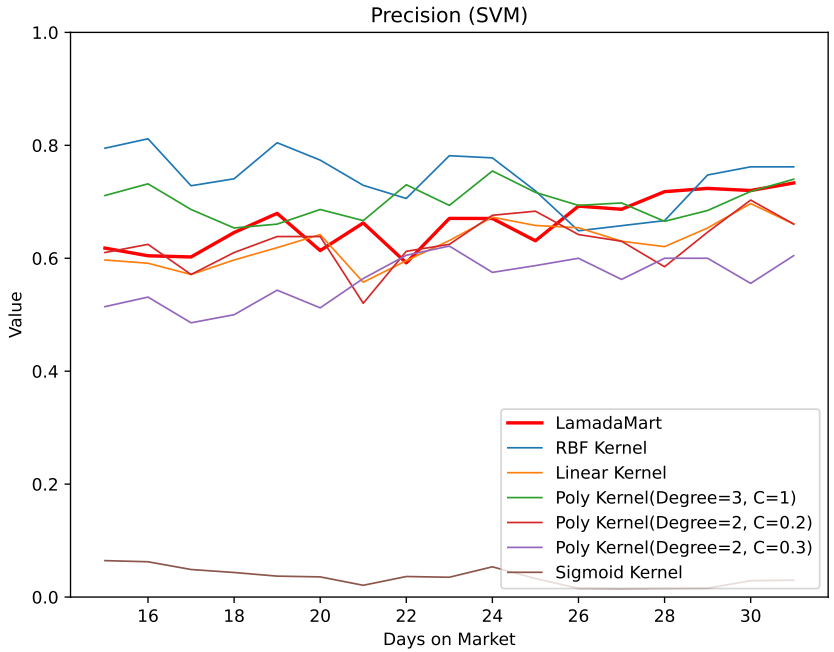
Prediction Algorithm

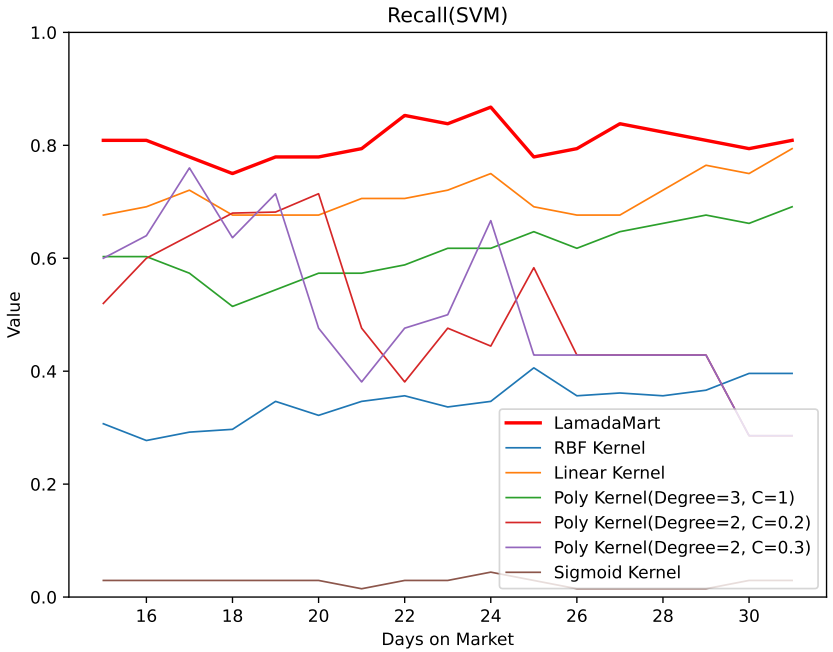
Table 2 **Precision and Recall Rates of 6 Different Models in Identification of Fast-selling Products**

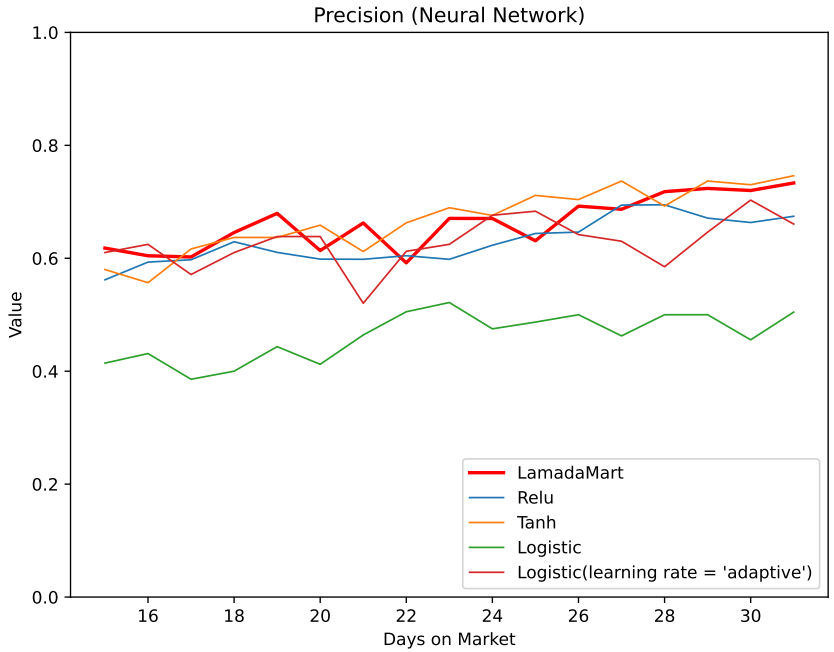
	Algorithm	14 Days	21 Days	1 Month	2 Month	3 Month
Precision	LambdaMart	**0.635** ⋆	0.663	**0.733** ⋆	**0.841** †	**0.895** †
	MLR	0.632	0.569	0.604	0.759	**0.770** ⋆
	SVM	**0.720** †	**0.729** †	**0.762** †	0.530	0.573
	XGB	0.594	**0.680** ⋆	0.717	0.747	0.706
	RF	0.557	0.576	0.604	0.632	0.661
	NN	0.611	0.648	0.724	**0.766** ⋆	0.732
Recall	LambdaMart	**0.794** †	**0.794** †	**0.809** †	**0.779** †	**0.784** †
	MLR	0.629	**0.634** ⋆	**0.673** ⋆	**0.733** ⋆	**0.776** ⋆
	SVM	0.267	0.347	0.396	0.614	0.574
	XGB	0.564	0.515	0.351	**0.733** ⋆	0.713
	RF	**0.728** ⋆	0.564	0.619	0.535	0.589
	NN	0.708	0.520	0.559	0.728	0.721

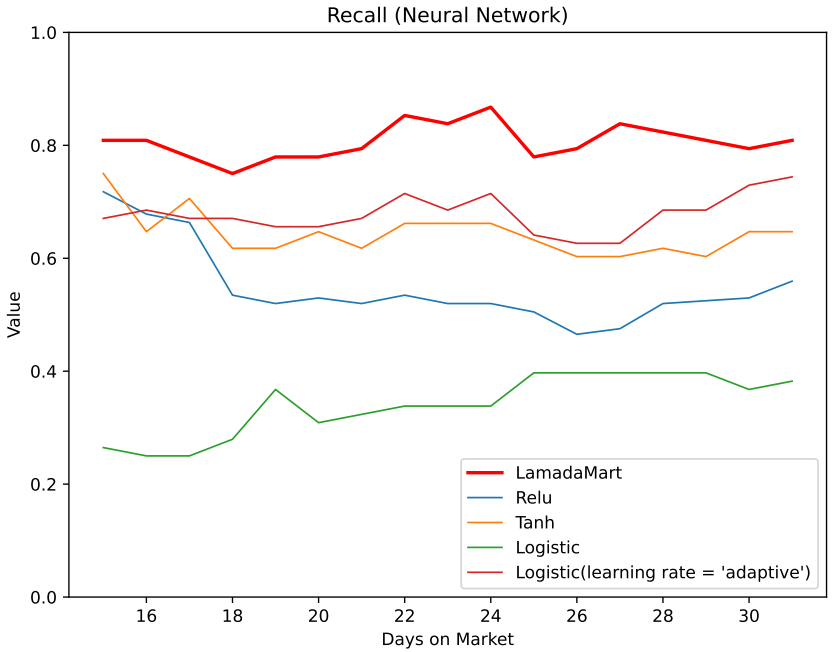
Notes. The red numbers († sign) represent the largest value among seven models and the blue numbers (⋆ sign) represent the second largest value.

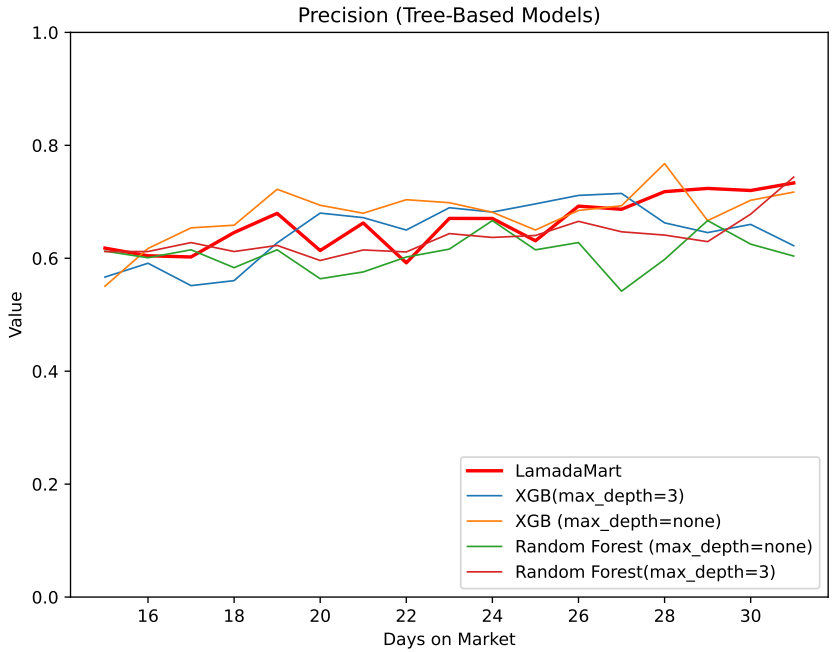
Figure: Comparision with other machine learning methods

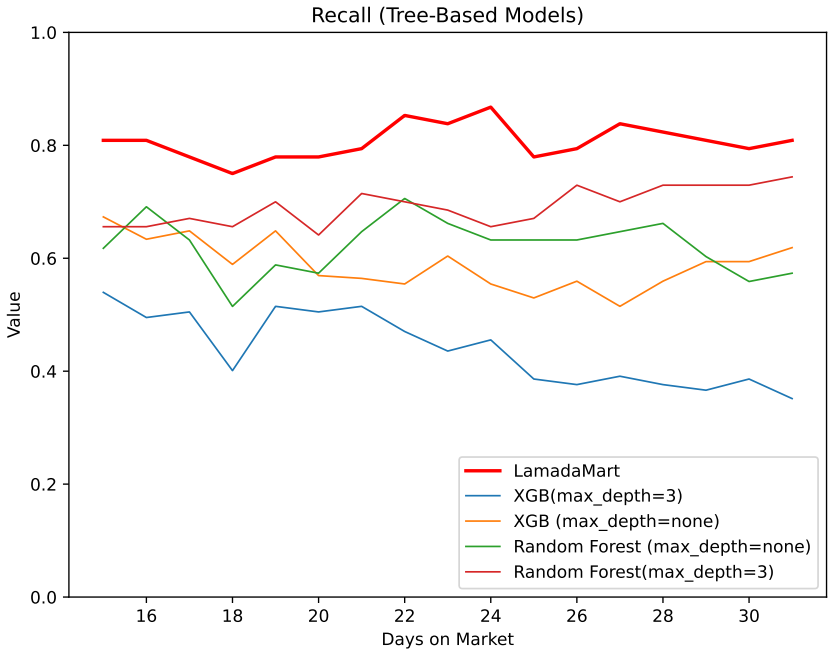

Precision (SVM)
Value
Days on Market
LamadaMart
RBF Kernel
Linear Kernel
Poly Kernel(Degree=3, C=1)
Poly Kernel(Degree=2, C=0.2)
Poly Kernel(Degree=2, C=0.3)
Sigmoid Kernel


Recall(SVM)
Value
Days on Market
LamadaMart
RBF Kernel
Linear Kernel
Poly Kernel(Degree=3, C=1)
Poly Kernel(Degree=2, C=0.2)
Poly Kernel(Degree=2, C=0.3)
Sigmoid Kernel


Precision (Neural Network)
Value
Days on Market
LamadaMart
Relu
Tanh
Logistic
Logistic(learning rate = 'adaptive')


Recall (Neural Network)
Value
Days on Market
LamadaMart
Relu
Tanh
Logistic
Logistic(learning rate = 'adaptive')


Precision (Tree-Based Models)
Value
Days on Market
LamadaMart
XGB(max_depth=3)
XGB (max_depth=none)
Random Forest (max_depth=none)
Random Forest(max_depth=3)


Recall (Tree-Based Models)
Value
Days on Market
LamadaMart
XGB(max_depth=3)
XGB (max_depth=none)
Random Forest (max_depth=none)
Random Forest(max_depth=3)

Model Evaluation

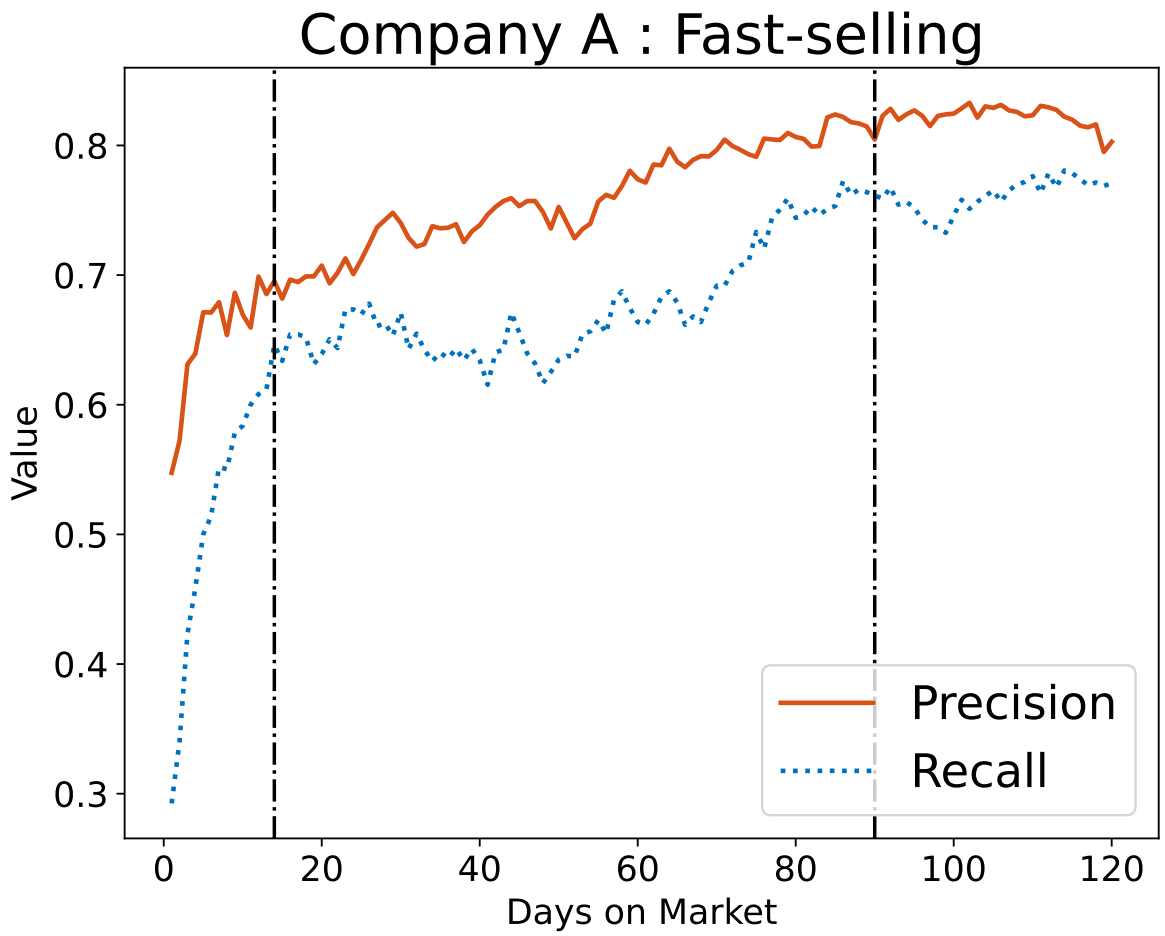
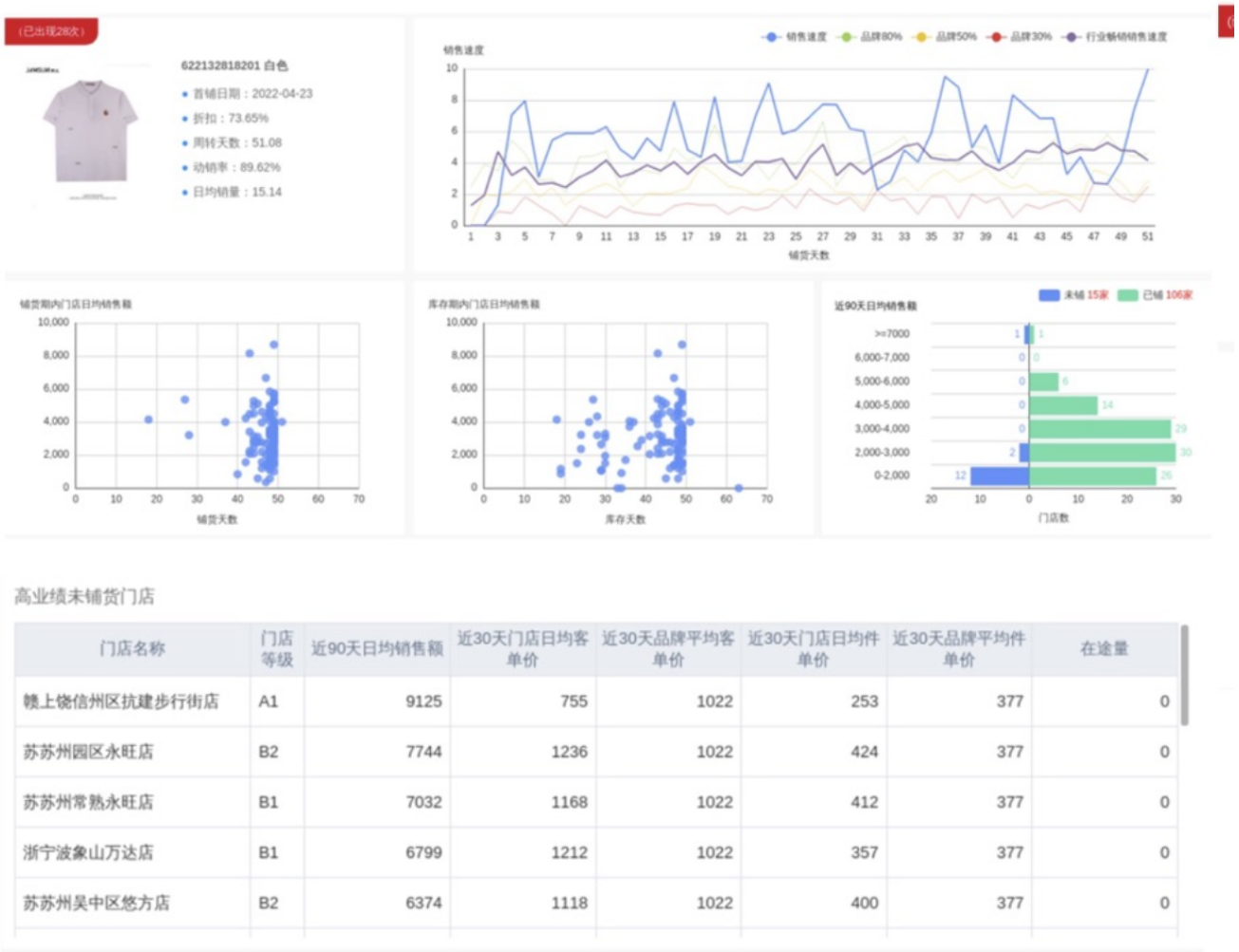


Figure: Precision and Recall Rates at Different Days on Market

Implemention



门店名称	门店等级	近90天日均销售额	近30天门店日均客单价	近30天品牌平均客单价	近30天门店日均件单价	近30天品牌平均件单价	在途量
赣上饶信州区抗建步行街店	A1	9125	755	1022	253	377	0
苏苏州园区永旺店	B2	7744	1236	1022	424	377	0
苏苏州常熟永旺店	B1	7032	1168	1022	412	377	0
浙宁波象山万达店	B1	6799	1212	1022	357	377	0
苏苏州吴中区悠方店	B2	6374	1118	1022	400	377	0

Figure: UI of develped software module

Decrease in the number of orders.

The number of products with more than one order decreases by 66% compared to the same period in 2021. The maximum number of orders placed on one product drops from 7 to 5.

Increase in sell-through rate.

Increase in sell-through rate from 50.88% to 85.53% at the end of the season in 2021 and a 74.4% reduction in disposal costs.

Increase in sales volume.(assumption)

If the sales managers place orders completely according to the model's predictions, the sales volume can be improved by 39,558 units, accounting for 5.9% of actual total sale volume.

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Model construction

Notations

p: Product Pricing
c: Product cost
T: Maximum duration of the selling season
S_h: Average weekly store sales of fast-selling products.
S_l: Average weekly store sales of slow-selling products.
$P(t)$: Accuracy function of model identification

Profit Function

$$\begin{aligned} Profit(t) = & P(t)[n \cdot S_h \cdot (T-t) \cdot (p-c)] \\ & + [1-P(t)][n \cdot S_l \cdot (T-t) \cdot (p-c) \\ & - n \cdot (S_h - S_l) \cdot (T-t) \cdot c] \end{aligned} \tag{3}$$

Simple case: $P(t) = a * t + b$

$$
\begin{aligned}
Profit(t) &= n \cdot [k_1 \cdot P(t) \cdot (T - t) + k_2 \cdot t] - k_2 * T \\
k_1 &= p \cdot (S_h - S_l) \\
k_2 &= c \cdot S_h - p \cdot S_l
\end{aligned}
$$

Profit function is convex in t.

$$
\begin{aligned}
t^* &= [\frac{c}{p} + (\frac{c}{p} - 1) \cdot \frac{S_l}{S_h - S_l}] \cdot \frac{1}{2a} - \frac{b}{2a} + \frac{T}{2} \\
&= \frac{k_2}{k_1} \cdot \frac{1}{2a} - \frac{b}{2a} + \frac{T}{2}
\end{aligned}
$$

P(t) is piecewise linear with N segments

$$P(t) = \begin{cases} a_1 \cdot t + b_1, & t \leq \tau_0 \\ a_2 \cdot t + b_2, & \tau_0 < t < \tau_1 \\ \dots & \\ a_N \cdot t + b_N, & \tau_{N-2} < t < T \\ a_1 \cdot \tau_0 + b_1 = a_2 \cdot \tau_0 + b_2 & \\ a_2 \cdot \tau_1 + b_2 = a_3 \cdot \tau_1 + b_3 & \\ \dots & \\ a_{N-1} \cdot \tau_{N-2} + b_{N-1} = a_N \cdot \tau_{N-2} + b_N & \\ 0 < a_2 < a_1 < \dots < a_N < 1 & \end{cases} \tag{4}$$

$$\alpha_n^* = \frac{k_2}{k_1} \cdot \frac{1}{2a_n} - \frac{b_n}{2a_n} + \frac{T}{2}$$

Lemma 1:

$\forall n \in \{2, \ldots, N\}, m \in \{0, \ldots, N-2\}$
When $\alpha_n^* > 0$, then there must be $\alpha_{n-1}^* > 0$
When $\alpha_n^* > \tau_m$, then there must be $\alpha_{n-1}^* > \tau_m$

$\forall n \in \{1, \ldots, N-1\}, m \in \{0, \ldots, N-2\}$
When $\alpha_n^* < 0$, then there must be $\alpha_{n+1}^* < 0$
When $\alpha_n^* < \tau_m$, then there must be $\alpha_{n+1}^* < \tau_m$

Lemma 2:

When $\frac{k_2}{k_1} < \frac{a_{n-1} \cdot b_n - a_n \cdot b_{n-1}}{a_{n-1} - a_n}, n = 2, 3, \ldots, N$, then there must be $\alpha_N^* < \alpha_{N-1}^* < \ldots < \alpha_2^* < \alpha_1^*$

Theorem

When risk is low: $\frac{k_2}{k_1} <= \frac{a_{n-1} \cdot b_n - a_n \cdot b_{n-1}}{a_{n-1} - a_n}, n = 2, 3, \ldots, N$

$$t^* = \begin{cases} 0, & \frac{k_2}{k_1} < b_1 - a_1 \cdot T \\ \alpha_1^*, & b_1 - a_1 \cdot T < \frac{k_2}{k_1} < b_1 - a_1 \cdot (T - 2\tau_0) \\ \alpha_0, & b_1 - a_1 \cdot (T - 2\tau_0) < \frac{k_2}{k_1} < b_2 - a_2 \cdot (T - 2\tau_0) \\ \alpha_2^*, & b_2 - a_2 \cdot (T - 2\tau_0) < \frac{k_2}{k_1} < b_3 - a_3 \cdot (T - 2\tau_1) \\ \ldots\ldots\ldots & \\ \alpha_n^*, & b_n - a_n \cdot (T - 2\tau_{n-2}) < \frac{k_2}{k_1} < b_n - a_n \cdot (T - 2\tau_{n-1}) \\ \alpha_{n-1}, & b_n - a_n \cdot (T - 2\tau_{n-1}) < \frac{k_2}{k_1} < b_{n+1} - a_{n+1} \cdot (T - 2\tau_{n-1}) \\ \ldots\ldots\ldots & \\ \alpha_N^*, & b_N - a_N \cdot (T - 2\tau_{N-2}) < \frac{k_2}{k_1} < 0 \end{cases}$$

Theorem

When risk is high: $\frac{k_2}{k_1} > \frac{a_{n-1} \cdot b_n - a_n \cdot b_{n-1}}{a_{n-1} - a_n}, n = 2, 3, \ldots, N$

$$t^* = \begin{cases} 0, & \frac{k_2}{k_1} < b_1 - a_1 \cdot T \\ \alpha_1^*, & b_1 - a_1 \cdot T < \frac{k_2}{k_1} < b_1 - a_1 \cdot (T - 2\tau_0) \\ \alpha_2^*, & b_2 - a_2 \cdot (T - 2\tau_0) < \frac{k_2}{k_1} < b_2 - a_2 \cdot (T - 2\tau_1) \\ \cdots\cdots\cdots & \\ \alpha_n^*, & b_n - a_n \cdot (T - 2\tau_{n-2}) < \frac{k_2}{k_1} < b_n - a_n \cdot (T - 2\tau_{n-1}) \\ \cdots\cdots\cdots & \\ \alpha_N^*, & b_N - a_N \cdot (T - 2\tau_{N-2}) < \frac{k_2}{k_1} < b_N - a_N \cdot (T - 2\tau_{N-1}) \\ T, & b_N - a_N \cdot (T - 2\tau_{N-1}) < \frac{k_2}{k_1} \end{cases}$$

Proof

Risk is low

when $b_n - a_n \cdot (T - 2\tau_{n-2}) < \frac{k_2}{k_1} < b_n - a_n \cdot (T - 2\tau_{n-1})$
$\rightarrow \tau_{n-2} < \alpha_n^* < \tau_{n-1}$
$\rightarrow$ profit function increases monotonically in (τ_{n-2}, α_n^*) and decreases monotonically in (α_n^*, τ_{n-1})

From Lemma2 $\quad (\alpha_N^* < \alpha_{N-1}^* < \ldots < \alpha_2^* < \alpha_1^*)$
$\alpha_n^* > \tau_{n-2} \rightarrow \alpha_1^* > \alpha_2^* > \ldots > \alpha_{n-1}^* > \tau_{n-2}$
$\rightarrow$ profit function increases monotonically in $(0, \tau_{n-2})$
$\alpha_n^* < \tau_{n-1} \rightarrow \alpha_N^* < \alpha_{N-1}^* < \ldots < \alpha_{n+1}^* < \tau_{n-1}$
$\rightarrow$ profit function decreases monotonically in (τ_{n-1}, T)

Proof

when $b_n - a_n \cdot (T - 2\tau_{n-1}) < \frac{k_2}{k_1} < b_{n+1} - a_{n+1} \cdot (T - 2\tau_{n-1})$
$\rightarrow \alpha_n^* > \tau_{n-1}, \alpha_{n+1}^* < \tau_{n-1}$

From Lemma2 $(\alpha_N^* < \alpha_{N-1}^* < \ldots < \alpha_2^* < \alpha_1^*)$
$\alpha_n^* > \tau_{n-1} \rightarrow \alpha_1^* > \alpha_2^* > \ldots > \alpha_{n-1}^* > \tau_{n-1} \rightarrow$ the profit function increases monotonically $(0, \tau_{n-1})$

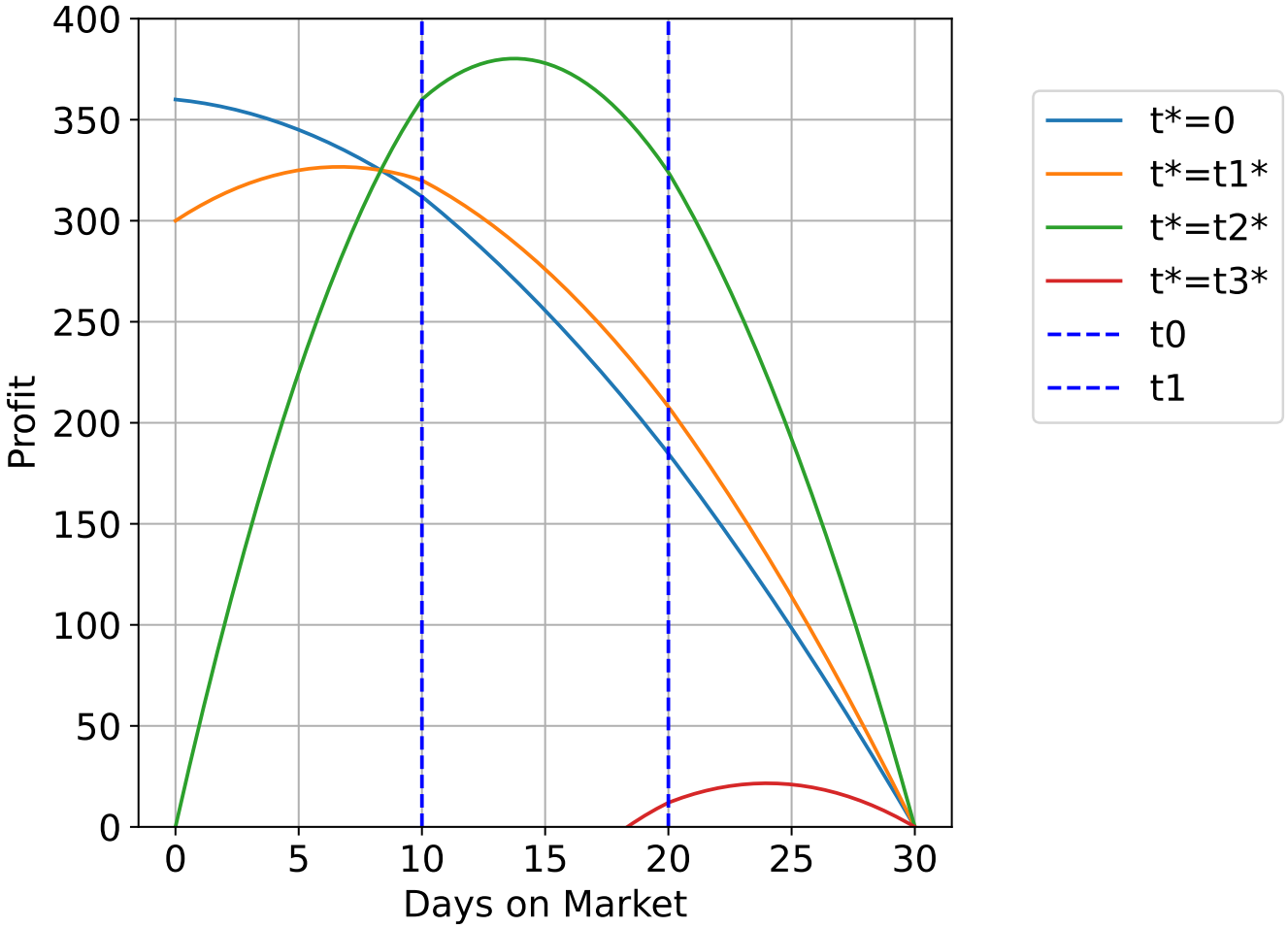
From Lemma1:
$\alpha_N^* < \alpha_{N-1}^* < \ldots < \alpha_{n+1}^* < \tau_{n-1} \rightarrow$ the profit function decreases monotonically (τ_{n-1}, T)

Proof

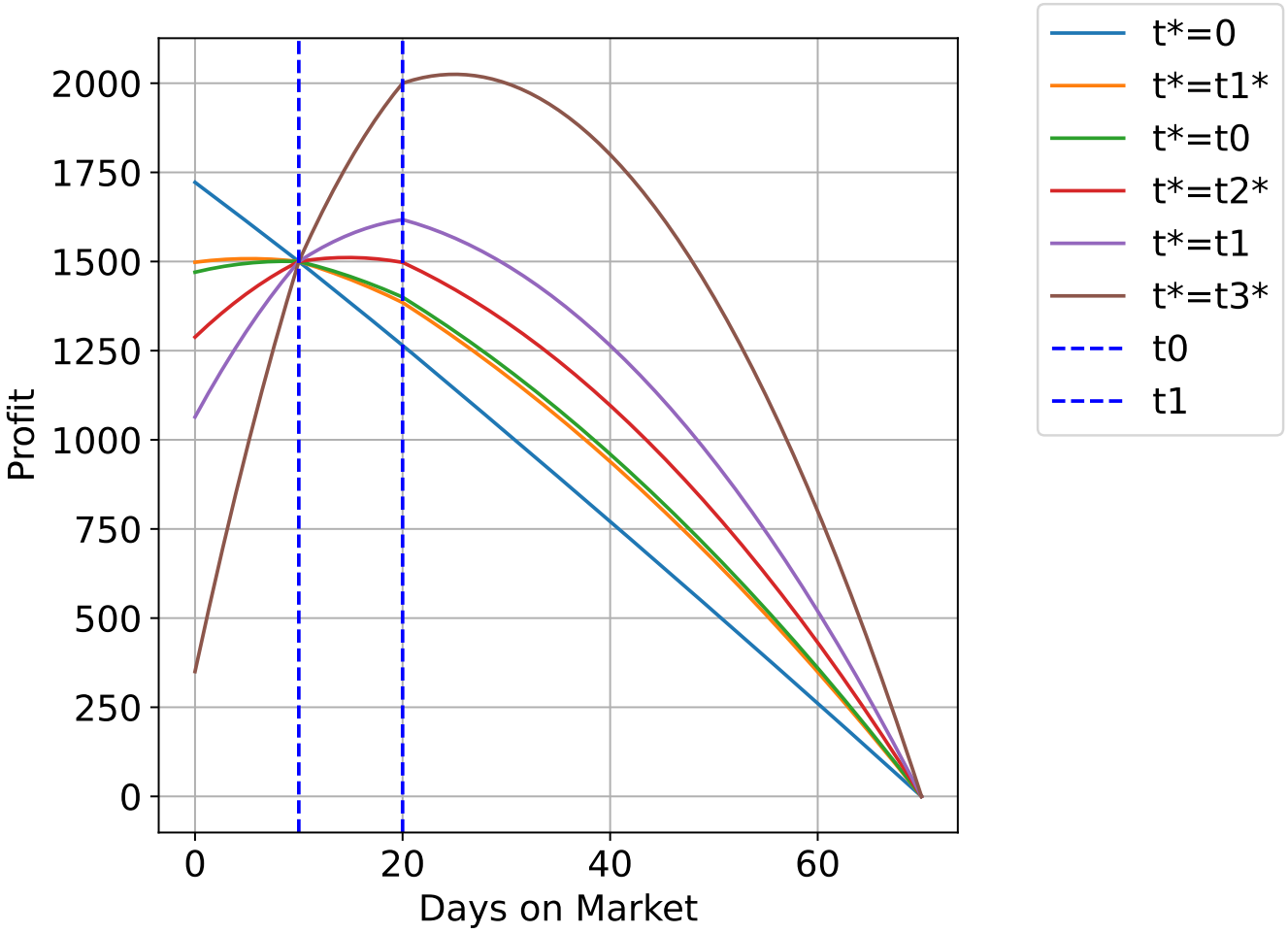
Risk is high

when $b_n - a_n \cdot (T - 2\tau_{n-1}) < \frac{k_2}{k_1} < b_{n+1} - a_{n+1} \cdot (T - 2\tau_{n-1})$
$\rightarrow \tau_{n-2} < \alpha_n^* < \tau_{n-1}$
$\tau_{n-2} < \alpha_n^* < \tau_{n-1}$,
$\rightarrow$ the profit function increases monotonically in $(\tau_{n-2}, \alpha_n^*)$the profit function decreases monotonically in (α_n^*, τ_{n-1})
From Lemma1
$\alpha_n^* < \tau_{n-1} \rightarrow \alpha_m^* < \tau_{n-1}, m \in \{n+1, n+2, \ldots, N\}$
$\rightarrow$ the profit function decreases monotonically in (τ_{n-1}, T)
From Lemma1
$\alpha_n^* > \tau_{n-2} \rightarrow \alpha_k^* < \tau_{n-2}, k \in \{1, 2, \ldots, n-1\}$
$\rightarrow$ the profit function increases monotonically in $(0, \tau_{n-2})$

Numerical Results- Small risk



Numerical Results- Big risk



Numerical Results

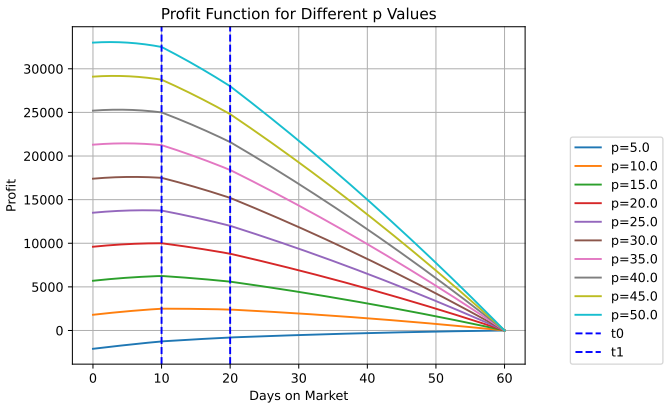


Figure: Price

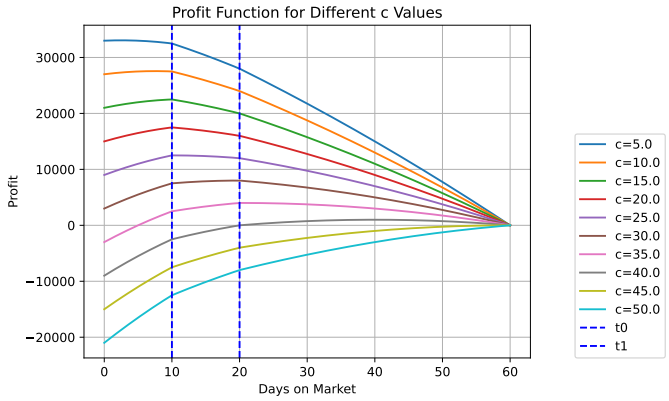


Figure: Cost

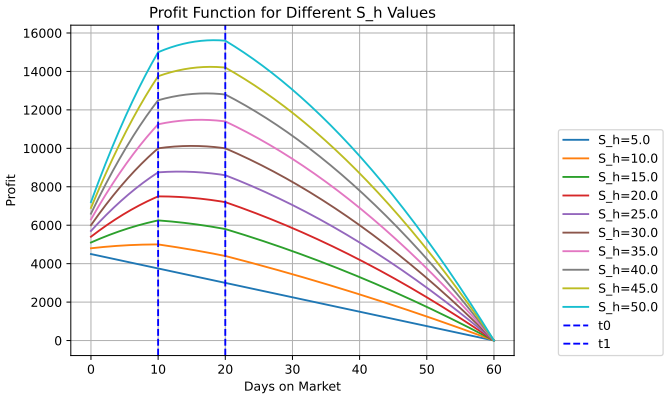


Figure: Sales High

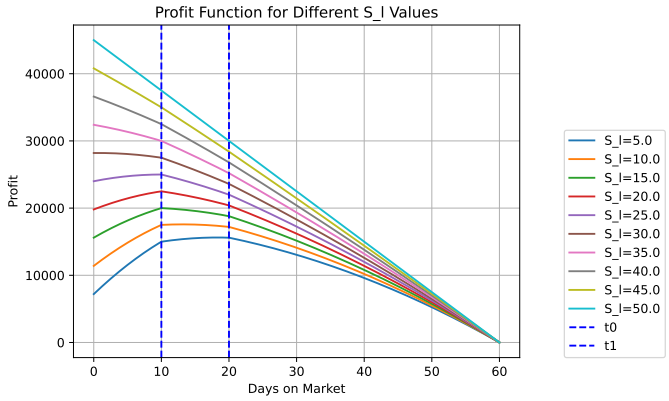


Figure: Sales Low

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Conclusion

Thesis Overview

The thesis comprises two integral studies. The first study pioneered a revolutionary product popularity classification model, while the second study constructed a mathematical model to ascertain the optimal reorder time.

Study 1: Product Popularity Classification

- Introduces a novel approach to identify fast-selling apparel products in the early stages of the selling season using machine learning techniques.
- Proposes the AW Sales metric to mitigate biases inherent in traditional metrics.
- Constructs innovative features, such as the longest increasing subsequence, to capture product sales trends and abnormal sales volumes.

- Utilizes the LambdaMart ranking model for modeling product popularity.
- Demonstrates that the model identifies fast-selling products multiple days earlier than a rule-based approach, resulting in higher sales rates and increased profits.

Study 2: Reorder Time Optimization

- Presents a decision model for determining the time to reorder fast-selling products when accuracy increases over time.
- Utilizes piece-wise accuracy functions with multiple intervals.
- Find the exact solution for the profit function with multiple intervals.
- Provides a comprehensive decision-making tool for sales managers to enhance decision-making processes, ensuring optimal ordering times and maximizing profitability.

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